

Variable-flavor attitude reports without underspecification

Jéssica Mendes 

Georg-August-Universität Göttingen
jessica.vianamendes@uni-goettingen.de

Giovanni Roversi 

MIT
grovers@mit.edu

Abstract: We examine an apparent case of a variable-flavor attitude predicate, the verb *kā* in Äiwoo, which can receive ‘think’, ‘say’, or ‘want’ translations. We argue that this verb denotes linguistic production, which may be mental or physical (thus capturing its use in doxastic and assertive reports). Its use for bouletic reports arises from combining it with an overt priority modal (the prefix *nā-*) in the embedded clause, whose semantic contribution can be independently established from its use in unembedded contexts.

Keywords: *attitude predicates, variable flavor, Äiwoo, Austronesian*

1 Introduction

Arguably, a defining feature of attitude verbs is that they are lexically specified for flavor: *think* is always interpreted doxastically, *want* is always interpreted bouletically, regardless of the context of use. This flavor rigidity stands in contrast to modal auxiliaries, like *may*, whose flavor is largely determined by the context. Recent work on understudied languages has challenged this generalization. Candidates for variable-flavor attitudes have been described in Navajo (Bogal-Allbritten 2016, 2017), Koryak (Močnik 2025, Močnik & Abramovitz 2019) and Secwepemctsin (Oliver 2025, 2026). The standard reaction to these findings is to build a component of underspecification into the semantics of attitudes. Specific implementations differ on whether they assume attitude verbs denote predicates of underspecified mental events, or quantifiers over possible worlds with modal bases retrieved from the context.

In this paper we discuss a new case of what appears to be a variable-flavor attitude verb: in Äiwoo, the attitude verb *kā* can be interpreted as ‘think/say’ or ‘want’ depending on the syntax of its complement. The basic pattern is given below: when the embedded verb bears aspectual morphology, *kā* is translated as either ‘think’ or ‘say’ (1); when the embedded verb instead bears the modal prefix *nā-*, *kā* is typically translated as ‘want’ (2).

(1) With embedded *i-* PFV, *ki-* IPFV: *kā* = ‘think, say’

- a. *kā* Sam=*kā* [_{CP} buk enge i-gââ-no]
KÄ Sam=CV book this PFV-read.UV-1MIN
‘Sam {thinks/thought/says/said} that I read this book’

- b. *kä* Sam=*kä* [_{CP} buk enge *ki-gââ-no*]
KÄ Sam=*CV* book this *IPFV*-read.UV-1MIN
 ‘Sam {thinks/thought/says/said} that I am/was reading this book’

(2) **With embedded *nä*:- *kä* = ‘want’**

- kä* Sam=*kä* [_{CP} buk enge *nâ-gââ-no*]
KÄ Sam=*CV* book this *NÄ*-read.UV-1MIN
 ‘Sam wants/wanted me to read this book’; lit. ‘Sam want{s/ed} that I read this book’¹

Our analysis of these facts departs from the underspecification-based approach pursued in previous work. We propose that *kä* denotes an event of linguistic production, which may be mental or physical (i.e., ‘say to oneself’ vs. ‘say out loud’). Since English does not have a verb with quite the same meaning, translations oscillate between ‘think’ and ‘say’. This is in line with existing crosslinguistic work (Demirok, Özyıldız & Öztürk 2019, Goodhue & Shimoyama 2024, Özyıldız & Uegaki 2024) and also corroborated by language-internal evidence. Bouletic readings, on the other hand, come from the prefix *nä*-, which has the semantics of a priority necessity modal. This point is supported by analyzing the contribution of *nä*- across different constructions, including matrix and purpose clauses. The upshot is that *kä* is not underspecified in any way, and its apparent flavor variability can be reduced to imprecise mappings in translation and to the role of different complements in shaping the meaning of attitude reports.

More generally, we show that our proposal is not only able to capture the Äiwoo facts, but it also makes better typological predictions. Crosslinguistically, variable-flavor attitude reports only express a small subset of possible attitude flavors: doxastic, assertive, and bouletic – with the latter relying on material embedded in the complement clause. While approaches based on underspecification can be enriched to capture constraints within a single language, it is not clear how they could provide a non-stipulative explanation to the broader typological pattern.

2 Previous accounts of variable-flavor attitude reports

2.1 The empirical landscape

Cases of predicates giving rise to attitude reports with variable-flavor interpretations have been analyzed for at least three languages. Bogal-Allbritten (2016, 2017) gives an influential analysis of such a predicate, the Navajo verb *nízin*. Attitude reports with *nízin* may be translated as e.g. ‘think’, ‘want’, or ‘wish’. Depending on the morphology in the embedded clause, in certain cases an attitude report may simply be ambiguous, with multiple readings being possible (3). Alternatively, overt material in the embedded clause may force one reading over the other: adding *sha’shin* ‘probably’ to (3) forces a ‘think’ reading (4a); similarly, removing the future marking from the embedded verb has the same effect of excluding the bouletic reading (4b). Other particles, not shown here, result in a ‘wish’ reading.

¹Äiwoo is largely a tenseless language, with the sole potential exception of the “future” particle =*Caa* (see later examples). Going forward, we only include a single tense in the English translations, for simplicity.

- (3) Alice [nahodoołtʃi] **nízin**. (Bogal-Allbritten 2017: ex. 3)
 Alice 3SG.rain.FUT 3SG.**nízin**.IPFV
 ✓ ‘Alice **thinks** it will rain.’; ✓ ‘Alice **wants** it to rain.’
- (4) a. [nahodoołtʃi sha’shin] **nisin**. (elaborated from Bogal-Allbritten 2017: ex. 2)
 3SG.rain.FUT probably 1SG.**nízin**.IPFV
 ✓ ‘I **think** it will probably rain.’; ✗ ‘I **want** it to rain.’
 b. Mary [nahaltin] **nízin**.
 Mary 3SG.rain.IPFV 3SG.**nízin**.IPFV
 ✓ ‘Mary **thinks** it is raining.’; ✗ ‘Mary **wants** it to rain.’

A somewhat similar pattern arises in Koryak (Močnik & Abramovitz 2019, Močnik 2025). Here, attitude reports with the verb *ivək* can receive a variety of interpretations: assertive, doxastic, and with variable quantificational force (5a). However, if the embedded verb bears counterfactual morphology (5b), the only available reading for *ivək* is ‘wish’.

- (5) a. mełło **kivəŋ** (əno) kumuqetəŋ (Močnik & Abramovitz 2019: 495)
 Melljo **ivək**.PRS.3SG that rain.PRS.3SG
 ‘Melljo {says, thinks, allows for the possibility, hopes, fears} that it’s raining.’
 b. mełło **kivəŋ** (iwke) nəʔəmuqetən
 Melljo **ivək**.PRS.3SG if.only rain.CF.3SG
 ‘Melljo wishes it would rain’

Another case is that of the attitude verb *tsut* in Secwepemctsin (Oliver 2025, 2026). Differently from Navajo and Koryak, in this language the different readings allowed by this predicate are consistently distinguished by overt morphology in the embedded clause: the complementizer *tek* only gives rise to assertive and doxastic readings (6a), while the purposive complementizer *e* only gives rise to bouletic readings (6b).

- (6) a. **tsut**-Ø lu? [**tek** s-geyp-s-t-Ø-és re=Julia]
 ATT=3SBJ REM COMP.OBL NMLZ-anger-CAUS-TR-3OBJ-3ERG DET=Julia
 ‘She **said/thought** that she made Julia mad’
 b. **tsú**(ts)t=ken [e s-tʔé(ʔ)y-n[-t]-Ø-n re=Julia]
 ATT<1SG>=1SG.SBJ PRP NMLZ-meet<1SG>-CTR-TR-3OBJ-1SG.ERG DET=Julia
 ‘I **want/intend** to meet Julia’ (data from Oliver’s 2025 abstract)

2.2 The analyses

To account for the behavior of *nízin* in Navajo, Bogal-Allbritten (2016, 2017) follows earlier proposals by Kratzer (2006, 2013) and Moulton (2009) in departing from the standard Hintikka model (Hintikka 1969), where the flavor of an attitude predicate (doxastic, assertive, bouletic, etc.) is lexically encoded as part of the denotation of the predicate itself. In this model, *nízin* is a relatively “empty” predicate of situations describing mental attitudes:

- (7) $\llbracket nízin \rrbracket^c = \lambda s \lambda w. \text{MENTAL-ATTITUDE}(s)(w)$

(8) a. $\llbracket \text{PRIORITY} \rrbracket^c = \lambda p \lambda s \lambda w. \forall w' \in \text{PRI}(s)(w). p(w')$
b. $\llbracket sha'shin / \text{ASSERT} \rrbracket^c = \lambda p \lambda s \lambda w. \forall w' \in \text{EPIS}(s)(w). p(w')$

(9) a. $\llbracket tsut \rrbracket = \lambda s \lambda w. s$ is a mental or verbal attitude in w
 b. $\llbracket tek/\emptyset \rrbracket = \lambda p \lambda s \lambda w. \forall w' \in \text{EPIS}_{\text{content}}(s)(w) : p(w')$
 (where s is a mental state or speech event)
 c. $\llbracket e \rrbracket = \lambda p \lambda s \lambda w. \forall w' \in \text{ROOT}(s)(w) : p(w')$

(10) $\llbracket iv\partial k \rrbracket^{c, g, \langle \langle a, v, S \rangle, s', w \rangle} (i)(C)(p)(x)$ (Močnik 2025: ex. 220)

- a. is defined only if
 - i. $i(x)(w) = \mathcal{B}_w^x$ **or** $i(x)(w) = \mathcal{L}_w^x$, &
 - ii. $C = \{f \mid f(i(x)(w)) = i(x)(w)\}$ **or** $C = \{f \mid f(i(x)(w)) \subseteq i(x)(w) \wedge f(i(x)(w)) \neq \emptyset\}$
- b. if defined, $\llbracket iv\partial k \rrbracket^{c, g, \langle \langle a, v, S \rangle, s', w \rangle} (i)(C)(p)(x)$ is true iff
 $\exists f \in C : \forall w' \in f(i(x)(w)) [p(\langle \langle x, w, i \rangle, f(i(x)(w)), w' \rangle) = 1]$

²This analysis very closely resembles the earlier proposal in Močnik & Abramovitz (2019).

4

Although this accounts models the precise range of meanings that *ivək* can have, it faces a potential overgeneration challenge. As far as we can tell, there is nothing principled restricting the disjunction in the presupposition (10a-i) to only range over doxastic and assertive modal bases. Absent any such restrictions, simply by modulating which modal base(s) are made available by this kind of presupposition, we could expect to find verbs in other languages whose range of meanings is different; such a hypothetical verb might alternate between e.g. ‘say’, ‘expect’, ‘hate’, and ‘imagine’. However, the range of interpretations that are found in cases of variable-flavor attitude reports cross-linguistically seems to be constrained: essentially, we only ever find verbs of this kind meaning (some subset of) ‘say’, ‘think’, and ‘want’ (or other doxastic/bouletic variants, like ‘hope’ or ‘wish’), but never really anything else. After a brief detour with some necessary background about Äiwoo (§3), we offer an analysis of the Äiwoo verb *kä* that captures this typologically restricted pattern (§§4–5).

3 Short background on Äiwoo

Äiwoo is a severely underresearched Oceanic (< Austronesian) language spoken by about 9 000 people in the Solomon Islands (Næss 2006, et seq.). The large part of the available literature is descriptive and functional/typological in nature (e.g. Næss 2015, et seq.). The only formal/generative work on Äiwoo focus largely on its syntax and morphology (e.g. Roversi 2020, 2025a,b). This paper is the first formal semantic study of any aspect of Äiwoo grammar. All uncited data comes from the second listed author’s ongoing fieldwork, carried out remotely since 2023 with one native speaker consultant, Luke Gitakulu. Other consulted data sources are a natural speech corpus collected by Åshild Næss over several fieldwork trips (2004–2015), and an in-progress Bible translation.

A few remarks will be useful for the reader. First, Äiwoo has a minimal/augmented number system (Næss 2006). These two categories can be thought of as loosely corresponding to singular and plural respectively; see e.g. Harbour (2016). Second, the language has an Austronesian voice system (Næss 2015), contrasting multiple transitive voices. Particularly, Undergoer Voice (UV) is the *de facto* default choice for transitive clauses, and it has OVS word order; the reader should not think of this as exceptional. For more details about Äiwoo syntax, see Roversi (2025a). The attitude verb *kä*, the main focus of this paper, is always in Circumstantial Voice (CV), a sort of applicative construction marked by the clitic particle =Cä⁴. The post-verbal position for the subject of *kä* (i.e., the attitude holder) is the expected one in CV; cf. (1)–(2). We assume that this applicative is necessary to integrate the embedded CP into the argument structure of the attitude verb, but at the moment we don’t have much more to say about this requirement. As a final note, both the attitude verb *kä* and the modal prefix *nä-* undergo vowel harmony (*kä~kâ*, *nä~nâ-*) when followed by a back vowel.

4 Kä in assertive/doxastic reports

First, we will focus on the use of *kä* in assertive and doxastic reports, i.e., when it’s translated as ‘say’ or ‘think’. This is the case when the verb in the embedded clause shows aspectual

⁴The initial consonant varies depending on orthogonal morphological factors; see Roversi (2025a: 62–63).

morphology (*i*- PFV, *ki*- IPFV) and no modal morphology (11). The key question we aim to answer in this section is how these two readings – and crucially not others – come about.

- (11) *kä* Sam=*kä* [_{CP} buk enge {*i*/*ki*}-gââ-no]
 KÄ Sam=CV book this {PFV/IPFV}-read.UV-1MIN
 ‘Sam {thinks/says} that I read/am reading this book’

We will consider two possible approaches for the denotation of *kä*. The first one, which we call ‘Single Meaning’ approach, is that the ‘think’ and ‘say’ translations of *kä* only imperfectly reflect a single underlying meaning that happens not to be lexicalized in English. The second view, which we call “Multiple Meanings” approach, is that by itself *kä* can mean *either* ‘think’ or ‘say’. Such a meaning could be modeled along the lines proposed by Močnik (2025) for Koryak: although *ivək* itself is not ambiguous, its modal flavor comes from a variable at LF, which is contextually resolved to *either* an assertive or a doxastic modal base. In the rest of this section, we argue for a Single Meaning approach, based on language-internal and cross-linguistic evidence.

4.1 Arguments in favor of a Single Meaning approach

Argument 1: conjoining two subjects for one *kä* Consider the case of a sentence containing one single occurrence of *kä*, but multiple attitude holders. Under a Multiple Meanings approach, this single *kä* should only be able to combine with either a doxastic or an assertive modal base variable, thus giving rise to only one of the two readings for both attitude holders.

Under a Single Meaning approach, in contrast, the single instance of *kä* has one interpretation, which is simultaneously compatible with both ‘think’ and ‘say’ translations; given the right context, this could result in two different translations for the two attitude holders. Indeed, this is precisely what we see (12)⁵.

- (12) [Context: Alice is always much more careful than John. He is always speaking about the future as if he actually know’s what’s gonna happen, he’s always super confident. She is more moderate/cautious about this.]
 [Jon mo Alice]=kâ kä-i=lä [teuwâ ku-po=kaa]
 John and Alice=DIST KÄ-3AUG=CV rain IPFV-go=FUT
 Lit.: ‘[John and Alice]_i, they_i *kä* that it will rain’
 ✓ ‘Alice *thinks* that it will rain and John *says* that it will rain’

Argument 2: conjoining two *kä*’s Consider now the case of a sentence of the form [*kä* P] & [¬*kä* P]. Under a Multiple Meanings approach, the modal base variable of each occurrence of *kä* could in principle be resolved to different values – doxastic in the first occurrence, assertive in the second, and vice-versa. That would lead to a non-contradictory reading being possible. A Single Meaning approach, in contrast, straightforwardly predicts contradiction.

⁵In (12), the conjunctive subject ‘John and Alice’ is topicalized, and resumed by a 3AUG suffixal pronoun on *kä* (as indicated by the free translation). This ensures that we don’t have any kind of ellipsis of a second *kä*.

This is borne out: sentences of this shape are robustly judged to be contradictory, even when the context ensures that each occurrence of *kä* should receive a different translation (13)^{6,7}.

- (13) [Context: Jane is a nurse working at a hospital, and there is a new doctor working there. She has seen this new doctor make many mistakes. She thinks the last doctor was much better.]
- a. # *kä* Jen=*kä* [dokta ki-tepeu] mo **ba** *kä*-Øⁿ=**gu**=*nä* [inâ ki-tepeu]
KÄ Jane=CV doctor IPFV-stupid but **NEG** *KÄ*-3MIN=**NEG**=CV 3MIN IPFV-stupid
 Int.: “Jane *kä*_{think} that the doctor is stupid, but she *didn’t* *kä*_{say} that he is stupid”
- b. # **ba** *kä* Jen=**gu**=*nä* [dokta ki-tepeu] mo *kä*-Øⁿ=*nä* [inâ ki-tepeu]
NEG *KÄ* Jane=**NEG**=CV doctor IPFV-stupid but *KÄ*-3MIN=CV 3MIN IPFV-stupid
 Int.: “Jane *didn’t* *kä*_{say} that the doctor is stupid, but she *kä*_{think} that he is stupid”
 Comment for both (13a,b): “This sounds contradictory”

Further cross-linguistic evidence As we saw earlier, three of the attitude verbs that appear in variable-flavor attitude reports (Koryak *ivək*, Secwepemctsin *tsut*, Äiwoo *kä*) convey ‘think’/‘say’⁸. A few morphemes with ‘think’/‘say’ translations have been discussed in other corners of the literature on attitude verbs: *ya* in Laz (Demirok, Özyıldız & Öztürk 2019), *to* in Japanese (Goodhue & Shimoyama 2024), and *diye-* in Turkish (Özyıldız & Uegaki 2024); the data is omitted for brevity. Adopting terminology from Demirok, Özyıldız & Öztürk (2019), all morphemes above give rise to *linguistic production inferences*, which correspond to either **mental or physical utterances**⁹.

Adopting a Single Meaning approach, and committing to the view that events involving linguistic production (whether mental or physical) form a natural class, provides a natural explanation to why different languages lexicalize morphemes that can be translated as ‘think’ or ‘say’. Simultaneously, this also **constrains** the range of “different flavors” a single attitude verb can be expected to express (without the help of additional material in the embedded clause), to essentially just ‘think’ and ‘say’. This is a welcome result: as far as we know, there are no reports of verbs whose reading alternates between e.g. ‘think’ and ‘expect’, or ‘say’ and ‘oblige’. Under a Multiple Meanings approach, some additional work would need to be done to rule out these (hitherto) unattested possibilities.

4.2 Formal analysis

To capture the behavior of *kä* in assertive/doxastic reports, we adopt Demirok, Özyıldız & Öztürk’s (2019) proposal: *kä* denotes a predicate of events that includes both *thinkings* and

⁶Note that standard clausal negation in Äiwoo is bipartite: *ba...* =*gu*, like *ne... pas* in literary French.

⁷Adding more linguistic material to each conjunct might perhaps facilitate the parse where the two *kä*’s have different intended meaning, in a similar way to how #*John must_{epist.} be home, but he must_{deon.} be at the hospital* is hard to parse as anything else than contradictory, but e.g. *His car is in the driveway, so John must_{epist.} be home, but he must_{deon.} be at the hospital before his shift starts* is somewhat easier to accept. We have not tested this in Äiwoo.

⁸This is potentially also the case for Navajo *nízin*, although the examples are less clear (Bogal-Allbritten 2017).

⁹See also e.g. Driemel & Kouneli (2025), Major (2021), and Major, Sikuku & Diercks (2026) on how ‘say’ verbs can have considerably more abstract meaning than just denoting a prototypical speaking event.

sayings (14). To be clear, we are not saying that *kä* can mean *either* ‘think’ or ‘say’. The meaning we give *kä* is the union of both: it is always going to be an event of a kind that covers both saying and thinking (cf. the section above). English simply doesn’t have any single verb with this lexical semantics, so we must use a disjunction to describe it, but there is nothing inherently disjunctive in the meaning we assign *kä*.

$$(14) \quad \llbracket kä \rrbracket^w = \lambda e.e \leq w[S \cup T(e)]$$

where $\llbracket S \cup T \rrbracket = \llbracket say \rrbracket \cup \llbracket think \rrbracket = \{e \mid e \text{ is a saying or a thinking event}\}$

We assume *kä* composes with its complement through the mediation of a covert declarative complementizer $\emptyset_{\text{CONT}}$, which has the same semantics as e.g. English *that* (15a)¹⁰. The resulting predicate of events (15b) then combines with *kä* via Predicate Modification (Bogal-Allbritten 2016, 2017, Bondarenko 2022, Elliott 2020, Kratzer 2016, a.o.), resulting in (15c).

$$(15) \quad \begin{aligned} \text{a. } & \llbracket \emptyset_{\text{CONT}} \rrbracket^w = \lambda p_{st} \lambda e.e \leq w[\text{CONT}(e)(w) \subseteq \{w' \mid \llbracket p \rrbracket^{w'} = 1\}] \\ \text{b. } & \llbracket [\text{CP } \emptyset_{\text{CONT}} \text{ buk enge igââno}] \rrbracket^w = \\ & \quad \lambda e.e \leq w[\text{CONT}(e)(w) \subseteq \{w' \mid \text{I read this book in } w'\}] \\ \text{c. } & \llbracket kä \emptyset_{\text{CONT}} \text{ buk enge igââno} \rrbracket^w = \\ & \quad \lambda e.e \leq w[S \cup T(e) \wedge \text{CONT}(e)(w) \subseteq \{w' \mid \text{I read this book in } w'\}] \end{aligned}$$

The attitude holder is introduced with the help a Θ -head, and existential closure applies, yielding the following truth-conditions:

$$(16) \quad \llbracket kä \text{ Sam}=kä \emptyset_{\text{CONT}} \text{ buk enge igââno} \rrbracket^w = 1 \text{ iff } \exists e.e \leq w$$

$[\text{HOLDER}(e) = \text{Sam} \wedge S \cup T(e) \wedge \text{CONT}(e)(w) \subseteq \{w' \mid \text{I read this book in } w'\}]$

In prose: There is an event of thinking or saying whose holder is Sam, and the content of that event is that I read this book.

In the next section we show how this basic meaning of *kä* composes with *nä-* to give rise to bouletic readings.

5 Bouletic reports with *kä* and *nä-*

Recall our baseline example: when *kä* combines with a CP whose verb carries the prefix *nä-* instead of aspectual morphology, the resulting report receives a ‘want’ reading (17).

$$(17) \quad \begin{aligned} & \text{kä Sam}=kä [\text{CP buk enge nâ-gââ-no}] \\ & \text{KÄ Sam}=CV \quad \text{book this NÄ-read.UV-1MIN} \\ & \text{‘Sam wants me to read this book’; lit. ‘Sam wants that I read this book’} \end{aligned}$$

First (§5.1), we examine the semantic contribution of *nä-* by looking at its use in unembedded contexts, and we show it is a necessity priority modal. Then (§5.2), we offer an analysis of why composing *kä* with *nä-* results in attitude reports with bouletic interpretations.

¹⁰We assume it’s reasonable to posit the existence of a null complementizer of this kind in Äiwoo, given that declarative CP complements also combine with other attitude verbs than *kä* (e.g. *wagu* ‘tell’, *kää* ‘know’, *ngo* ‘hear’, etc.) without any overt complementizer.

5.1 Establishing that *nä-* is a priority modal

When *nä-* is used in a matrix clause, it is translated as a modal, expressing **deontic** (18), **bouletic** (19), and **teleological necessity** (20)¹¹. Following Portner (2009), we subsume these three modal flavors under the label “priority”.

- (18) [Context: in the Solomon Islands, the laws state that when you ride a motorbike, ...]
 helmet {✓*nä*/#*i*/#*ki*}-vitee-mu ngâ nuwotaa-mu
 helmet {*NÄ*/*PFV*/*IPFV*}-put.on.UV-2MIN on head-2MIN
 a. With ✓*nä-viteemu*: ‘You must wear a helmet’
 b. Comment on #*i-viteemu*: “This sounds like past tense, ‘you put on a helmet’”
 c. Comment on #*ki-viteemu*: “This sounds like, ‘you are wearing a helmet’. *Nä*-sounds like an order”
- (19) [Context: I want my son to grow up speaking Äiwoo, and not just Pijin or English. Of course there is no law that says so, but I think it would be best.]
 ginou {✓*nä*/#*i*/#*ki*}-eeu ngä Äiwoo
 son.1MIN {*NÄ*/*PFV*/*IPFV*}-speak in Äiwoo
 a. With ✓*nä-eeu*: ‘My son has to speak Äiwoo’
 b. Comment on #*i-eeu*: “[You could use this] after a speech: ‘My son spoke Äiwoo’”
 c. Comment on #*ki-eeu*: “This means ‘My son is speaking Äiwoo’”
- (20) [Context: I am in the Reef Islands, and I want to go to Santa Cruz (the main island of the area). The only way to get there is by boat, there are no flights right now.]
 mi-{*nâ*/*ki*}-lo-kä bote
 2MIN-{*NÄ*/*IPFV*}-take.AV-DIR3 boat
 a. With ✓*mi-nâ-lokä*: ‘You have to take a boat’
 b. With ✓*mi-ki-lokä*: ‘You take a boat’¹²

We can also see that *nä-* is not underspecified for flavor. For example, it cannot express epistemic necessity (21); this meaning is expressed by a plain, non-modalized clause.

- (21) [Context: Mary routinely has coffee at Lisa’s house everyday. Even if she’s sick, she doesn’t miss a day! It’s not obligatory for Mary; she just goes for coffee there all the time. It’s coffee time now, so...]
 ilâkâ Mary {#*nâ*/✓*ku*}-mo ngâ nuwopa tä Lisa
 there Mary {*NÄ*/*IPFV*}-stay in house poss Lisa
 Intended: ‘Mary is/must_{epist.} be at Lisa’s house’

¹¹In all earlier descriptive literature (Næss 2006 et seq.), *nä-* was labeled/glossed as “irrealis”, but this was never made more precise. The precise range of semantic contributions made by *nä-* shown here is a novel empirical discovery. The contexts for these examples are adapted from Vander Klok (2022).

¹²In the case of teleological necessity the non-modalized statement with *ki-* (20b) is also acceptable, unlike for deontic and bouletic necessity (18c)–(19c). We suspect this might be a type of generic statement, of the kind “to go to Berlin, you take a train”.

5.2 Formal analysis

To capture its uses across constructions, we give *nä*- the following lexical entry:

- (22) $\llbracket n\ddot{a}\rrbracket^w = \lambda p_{st} \lambda e. e \leq w[\text{BEST}_{g(e)}(\cap f(e)) \subseteq \{w' \mid \llbracket p \rrbracket^{w'} = 1\}]$
- a. $\cap f$ is a **circumstantial modal base** containing the worlds that are compatible with the circumstances of e
 - b. $g(e)$ is a **teleological, deontic, or bouletic ordering source** that orders the propositions in $\cap f(e)$ according to goals, rules or desires in e (i.e.: the priorities of the agent of e)

Our treatment of *nä*- makes it, by and large, a standard doubly-relative modal (Kratzer 1981, 2012). We only depart slightly from the Kratzerian tradition in letting *nä*- project its modal domain from an event, instead of a possible world¹³. This is going to be crucial to account for *nä*- in bouletic reports and purpose clauses. But before we get to these cases, let's briefly consider what happens when *nä*- is unembedded.

Our proposal is that, in matrix clauses, *nä*- combines with its prejacent via Intensional Functional Application, then has its event variable existentially closed:

- (23) a. $\llbracket \text{helmet } n\ddot{a}viteemu \ ng\hat{a} \ nuwotaamu \rrbracket^w = 1$ iff
 $\exists e. e \leq w[\text{BEST}_{g(e)}(\cap f(e)) \subseteq \{w' \mid \text{you wear a helmet in } w'\}]$
- b. **In prose:** There is an event e such that in all worlds w' that rank highest according to the priorities of the participants of e , you wear a helmet in w' .

That leads to rather vague truth conditions: (23a) is true as long as there is some event whose circumstances require that the addressee wear a helmet; as it is, we need to let the context specify which event it is. A more sophisticated alternative would involve adopting a framework in which event proforms are represented in the syntax (Alonso-Ovalle, Menéndez-Benito & Rubinstein 2024, Hacquard 2006, a.o.), and then developing a theory of how these proforms are bound. If we pursued this view we could assume, for instance, that in matrix clauses the event argument of *nä*- is bound by a speech event. We leave an exploration of this idea for future work¹⁴.

With this much in place, the behavior of *nä*- in embedded clauses falls out naturally. In attitude reports, after combining with its prejacent, *nä*- combines with *kä* via Predicate Modification. As a result, both *kä* and *nä*- are anchored to the same event:

- (24) a. $\llbracket buk \ eng \ n\hat{a}g\hat{a}\hat{a}no \rrbracket^w = \lambda e. e \leq w[\text{BEST}_{g(e)}(\cap f(e)) \subseteq \{w' \mid \text{I read this book in } w'\}]$
- b. $\llbracket k\ddot{a} \ buk \ eng \ n\hat{a}g\hat{a}\hat{a}no \rrbracket^w =$
 $\lambda e. e \leq w[S \cup T(e) \wedge \text{BEST}_{g(e)}(\cap f(e)) \subseteq \{w' \mid \text{I read this book in } w'\}]$

¹³The idea that modal domains can be projected from events originally comes from Hacquard (2006).

¹⁴In Hacquard's original system, non-epistemic modals could be anchored to the speech event or to the νP event, depending on their height of attachment. When anchored to the speech event, non-epistemic modals acquire an addressee-oriented reading; when anchored to the νP event, modals have a subject-oriented reading. More empirical work needs to be done to determine if *nä*- is compatible with these two readings, and whether they correlate with syntactic differences.

The rest of the derivation proceeds straightforwardly, yielding the following truth conditions:

- (25) $\llbracket k\ddot{a} \text{ Sam}=k\ddot{a} \text{ buk enge } n\ddot{a}g\ddot{a}n\ddot{o} \rrbracket^w = 1$ iff
 $\exists e.e \leq w[\text{HOLDER}(e) = \text{Sam} \wedge S \cup T(e) \wedge \text{BEST}_{g(e)}(\cap f(e)) \subseteq \{w' \mid \text{I read this book in } w'\}]$
In prose: There is an event of thinking or saying whose holder is Sam, and in all the worlds that rank highest according to Sam’s priorities in e , I read this book.

The same analysis can be extended to other cases where $n\ddot{a}$ - is embedded, but not under $k\ddot{a}$; for instance, in adjunct purpose clauses (26):

- (26) elenge pedevalili i-li-momalee-eke [go nubole $n\ddot{a}$ -kili-mä-i]
 today children PFV-3AUG-wake.up-fast for taro $N\ddot{A}$ -dig.UV-DIR1-3AUG
 ‘Today the children woke up early to harvest the taro (for me)’
 (27) $\llbracket (26) \rrbracket^w = 1$ iff $\exists e.e \leq w[\text{PATIENT}(e) = \text{the children} \wedge \text{wake-up-early}(e) \wedge$
 $\text{BEST}_{g(e)}(\cap f(e)) \subseteq \{w' \mid \text{the children harvest the taro for me in } w'\}]$
In prose: There is an event e of the children waking up early, and in all worlds w' that rank highest according to the priorities of the participants of e , the children harvest the taro for me in w' .

To conclude this section, we briefly comment on two aspects of the syntax/semantics of $n\ddot{a}$ - that were not discussed above.

Complementary distribution of $n\ddot{a}$ - and i -/ki- As we saw throughout the paper, an Äi-woo verb may either carry the aspectual prefixes i -/ki- (perfective/imperfective), or it may carry $n\ddot{a}$ -, but not both. Nothing in our analysis so far captures these facts. For simplicity, we provisionally treat this as a shallower *morphological* incompatibility, not reflective of any deeper semantic significance: the Aspect head is morphologically deleted in the presence of $n\ddot{a}$ - (28a), and is otherwise exponed regularly as i -/ki- (28b). We leave it to future research to examine what the aspectual properties of $n\ddot{a}$ -clauses are, to establish whether the incompatibility between these prefixes is a deeper one.

- (28) a. $\text{Asp} \Leftrightarrow \emptyset / _ [\text{PRIORITY}]$ b. $\text{Asp}_{[\text{PFV}]} \Leftrightarrow i$ -
 $\text{Asp}_{[\text{IPFV}]} \Leftrightarrow ki$ -

The syntactic category of $n\ddot{a}$ - We have given $n\ddot{a}$ - the same semantic type as $\emptyset_{\text{CONT}}$, which we treated as a complementizer. Syntactically, $n\ddot{a}$ - could be treated either as head in the high T/Infl region (e.g. a modal/mood morpheme, like a **subjunctive**), or also as a “low” C head in a cartographic C-layer (Rizzi 1997), allowing for the presence of higher complementizers (e.g., *go* ‘for’ in (26)). Supporting the idea that $n\ddot{a}$ - is syntactically higher than the aspectual prefixes i -/ki-, $n\ddot{a}$ - shows allomorphic interactions with prefixal ϕ -agreement, which has been argued to be in the C region (Roversi 2025a); the two aspectual prefixes never do this.

Pushing this hypothesis seems like a promising route as it might also help explain the complementary distribution of $n\ddot{a}$ - and aspectual morphemes: there are many languages where e.g. the subjunctive paradigm collapses tense-aspect distinctions that are otherwise made

in indicative paradigms – e.g. Italian, Spanish, Kipsigis (Driemel & Kouneli 2025), a.o. Besides that, corpus evidence shows that *nä-* can also be used in *wish*-reports and X-marked conditionals, two environments that are typically associated with the subjunctive (von Fin-
tel & Iatridou 2023, Iatridou 2000, a.o.). Hence, pursuing this analogy between *nä-* and the subjunctive could shed light on the broader distribution of *nä-*.

6 Discussion

In this paper we have provided the first description and account of attitude reports with the verb *kä* in Äiwoo. Although *kä* might *prima facie* look like a variable-flavor attitude verb, we argued that its semantics is not underspecified in any way. *Kä* denotes events of linguistic production; hence, in the lack of better a better translation, it is rendered as either ‘think’ or ‘say’ in English. Bouletic readings come from material embedded in the complement clause, giving rise to a type of meaning alternation that is very well-attested crosslinguistically¹⁵.

The main advantage of our approach is that it precisely derives the meanings attested for *kä*, without allowing for additional, unattested readings. As we saw in §2, underspecification-based analyses can likewise prevent overgeneration by encoding constraints directly into the lexical entries of attitude verbs. While this strategy may be successful on a case-by-case basis, it risks making undesirable crosslinguistic predictions. The range of meanings attested for variable-flavor attitude reports is remarkably stable and limited across languages, and a theory of the phenomenon should capture this typological generalization.

Our work therefore opens some interesting avenues for future research. First, even though we have discussed data from four unrelated languages spoken on three continents, this is still a rather small sample: further empirical crosslinguistic work will be necessary to establish more robust generalizations, i.e., to verify whether the range of meanings expressed by variable-flavor attitude reports is really restricted to ‘think/say’ meanings and bouletic ones. We will then need to see if an analysis like the one we proposed for *kä* can be extended to other cases of variable-flavor attitude reports, or if some of them really do require underspecification.

If it does indeed turn out to be the case that all variable-flavor attitude reports can be treated how we treat *kä*, this would suggest that natural languages do not lexicalize genuine variable-flavor attitude verbs. Why should that be the case? At first glance, there seems to be no obvious reason why a lexical item expressing an attitude should have a more rigid semantics than, say, a modal such as *may* or *can*, which can receive different interpretations depending on the context. Understanding why the lexical semantics of attitude predicates should be subject to such a restriction could, therefore, provide insight into the division of labor between attitude predicates and other modal elements, and into the pressures that shape the repertoire of modal vocabulary across languages.

¹⁵Other instances of the same alternation include ‘convince that/convince to’, ‘promise that/promise to’, ‘tell that/tell to’, a.o. We thank Susi Wurmbrand (p.c.) for drawing our attention to this connection.

Abbreviations

1 = first person, 2 = second person, 3 = third person, ATT = attitude, AUG = augmented number, AV = actor voice, CAUS = causative, CF = counterfactual, COMP = complementizer, CTR = control, CV = circumstantial voice, DET = determiner, DIR = directional, DIST = distal, ERG = ergative, FUT = future, IPFV = imperfective, MIN = minimal number, NEG = negative, NMLZ = nominalizer, OBJ = object, OBL = oblique, PFV = perfective, POSS = possessive, PRP = purposive, PRS = present, REM = remote, SBJ = subject, SG = singular, TR = transitive, UV = undergoer voice.

Acknowledgements

Both authors contributed equally to the project, and the names are listed alphabetically. We are very grateful to audiences at GLOW48, SULA/Triple A, and MIT for insightful comments and questions. Special thanks to Scott AnderBois, Valentina Bianchi, Bruce Oliver, Ryan Walter Smith, Susi Wurmbrand and Michael Yoshitaka Erlewine for extended discussions. All remaining mistakes are ours. Jéssica Mendes is funded by an ERC grant (UNPAG, 101142366, 0.3030/101142366, PI: Prof. Hedde Zeijlstra). Views and opinions expressed are however ours only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency.

References

- Alonso-Ovalle, Luis, Paula Menéndez-Benito & Aynat Rubinstein. 2024. Eventive modal projection: the case of Spanish subjunctive relative clauses. *Natural Language Semantics* 32(2). 135–176.
- Anand, Pranav & Valentine Hacquard. 2013. Epistemics and attitudes. *Semantics and Pragmatics* 6(8). 1–59.
- Bogal-Allbritten, Elizabeth. 2016. *Decomposing attitudes: the view from Navajo*. UMass Amherst dissertation.
- Bogal-Allbritten, Elizabeth. 2017. *Basic pieces, complex meanings: Building attitudes in Navajo and beyond*. Handout from the 39th DGfS conference.
- Bondarenko, Tatiana. 2022. *Anatomy of an attitude*. MIT dissertation.
- Demirok, Ömer, Deniz Özyıldız & Balkız Öztürk. 2019. Complementizers with Attitude. *Proceedings of NELS* 49(1). 213–222.
- Driemel, Imke & Maria Kouneli. 2025. C-Agree Is Local Subject-Verb Agreement in Kipsigis. *Natural Language & Linguistic Theory* 43(1). 223–272.
- Elliott, Patrick D. 2020. *Elements of clausal embedding*. UCL dissertation.
- von Fintel, Kai & Sabine Iatridou. 2023. Prolegomena to a theory of X-marking. *Linguistics and Philosophy* 46(6). 1467–1510.
- Goodhue, Daniel & Junko Shimoyama. 2024. *Embedded Negative Polar Questions in Japanese: Explaining the Puzzling Distribution of Embedded Noncanonical Negation via the Speech Act Embedding Complementizer ‘to’*. Manuscript.
- Hacquard, Valentine. 2006. *Aspects of modality*. MIT dissertation.

- Harbour, Daniel. 2016. *Impossible Persons*. Cambridge, MA: MIT Press.
- Hintikka, Jaakko. 1969. Semantics for propositional attitudes. In *Models for modalities: Selected essays*, 87–111. Springer.
- Iatridou, Sabine. 2000. The grammatical ingredients of counterfactuality. *Linguistic Inquiry* 31(2). 231–270.
- Kratzer, Angelika. 1981. The notional category of modality. *Formal semantics: The essential readings*. 289–323.
- Kratzer, Angelika. 2006. *Decomposing attitude verbs*. Talk presented at the Hebrew University of Jerusalem.
- Kratzer, Angelika. 2012. *Modals and conditionals*. Oxford: Oxford University Press.
- Kratzer, Angelika. 2013. *Modality and the semantics of embedding*. Talk presented at the 20th Amsterdam Colloquium.
- Kratzer, Angelika. 2016. *Evidential moods in attitude and speech reports*. Talk given at the 1st Syncart Workshop.
- Major, Travis. 2021. *On the Nature of “Say” Complementation*. UCLA dissertation.
- Major, Travis, Justine Sikuku & Michael Diercks. 2026. Grammatically Citing Your Sources via “Say”-Complementation. *Natural Language & Linguistic Theory* 44(3). 1–40.
- Močnik, Maša. 2025. *Strange attitudes on top*. MIT dissertation.
- Močnik, Maša & Rafael Abramovitz. 2019. A variable-force variable-flavor attitude verb in Koryak. *Proceedings of the Amsterdam Colloquium* 22. 494–503.
- Moulton, Keir. 2009. *Natural selection and the syntax of clausal complementation*. UMass Amherst dissertation.
- Næss, Åshild. 2006. Bound Nominal Elements in Äiwoo (Reefs): A Reappraisal of the “Multiple Noun Class Systems”. *Oceanic Linguistics* 45(2). 269–296.
- Næss, Åshild. 2015. Voice at the Crossroads: Symmetrical Clause Alternations in Äiwoo, Reef Islands, Solomon Islands. *Oceanic Linguistics* 54(1). 270–307.
- Oliver, Bruce. 2025. *Attitude comes from the complement clause: Evidence from Secwepemctsin*. Talk presented at the 2025 CLA conference.
- Oliver, Bruce. 2026. *That’s too vague! Ambiguity avoidance with indeterminate attitude predicates*. Talk presented at the 2026 SULA–Triple A conference.
- Özyıldız, Deniz & Wataru Uegaki. 2024. *Clauses can be modifiers, or arguments: Evidence from alternations in factivity and answer-orientedness in Turkish and Japanese*. Ms., Universität Konstanz & University of Edinburgh. <https://ling.auf.net/lingbuzz/008187>.
- Portner, Paul. 2009. *Modality*. Oxford: Oxford University Press.
- Rizzi, Luigi. 1997. The Fine Structure of the Left Periphery. In Liliane Haegeman (ed.), *Elements of Grammar: Handbook in Generative Syntax*, 281–337. Dordrecht: Springer.
- Roversi, Giovanni. 2020. How to satisfy probes: person/number hierarchy effects in Äiwoo. *Proceedings of NELS* 50(3). 99–112.
- Roversi, Giovanni. 2025a. *Locality: a case study from Äiwoo*. MIT dissertation.
- Roversi, Giovanni. 2025b. Possession and syntactic categories: An argument from Äiwoo. *Natural Language & Linguistic Theory* 43(1). 385–420.
- Vander Klok, Jozina. 2022. Revised modal questionnaire for cross-linguistic use.
- Williamson, Gregor. 2021. *Worlds of the future: Modality & future licensing at the syntax-semantics interface*. UCL dissertation.